

Q.18 What is a torque converter? How does it work?

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Explain the construction and working of an epicyclic gearbox. List its applications and advantages over sliding mesh gearbox.

Q.20 Explain the constructional features and working of a multi plate clutch.

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Level 4, 2nd Sem / DVOC (Auto, Servicing, AMT)

Subject : Motor Vehicle Technology - I

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Multiple Choice questions. All questions are compulsory. (5x1=5)

Q.1 The distance between the front and rear axles is called:

- a) Wheel track b) Ground clearance
- c) Overhang d) Wheelbase

Q.2 Which of the following is used in Maruti cars for smooth gear shifting?

- a) Diaphragm clutch b) Fluid coupling
- c) Cone clutch d) Centrifugal clutch

Q.3 The function of the PTO (Power Take-Off) shaft in a tractor is to:

- a) Drive the wheels
- b) Control the steering
- c) Power external implements like rotavators
- d) Supply hydraulic pressure

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- Q.4 Universal joints are used to:
- a) Convert AC to DC
 - b) Allow angular movement in shafts
 - c) Align gears
 - d) Cool down the differential

- Q.5 Tyre retreading in the process of:
- a) Recycling rubber
 - b) Replacing the sidewall
 - c) Replacing worn-out tread with new rubber
 - d) Refilling air with nitrogen

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

- Q.6 _____ weight is the weight of the vehicle with all fluids but without passengers or cargo.
- Q.7 _____ coupling uses fluid to transmit torque without mechanical contact.
- Q.8 A gearbox changes the _____ and torque of a vehicle.

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- Q.9 The three-Quarter floating axle carries part of the vehicle _____ and torque.
- Q.10 Nitrogen in tyres maintains stable pressure because it is less affected by _____ changes.

SECTION-C

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

- Q.11 Describe the front-wheel drive and rear-wheel drive along with their advantages and disadvantages.
- Q.12 Explain the necessity and functions of a clutch in an automobile.
- Q.13 Explain the working of synchromesh with a diagram.
- Q.14 Define final drive. Name its main components and their purpose.
- Q.15 List different types of tyres used in automobiles. Mention features of any two types.
- Q.16 Explain the importance of tyre retreading and wheel balancing. How do these practice improve tyre performance and safety.
- Q.17 Why a slip in joint necessary to be used with propeller shaft?

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